Reading Tiramisu in Czech and English:
Robust Processing Speed Differences in Translation Equivalent Stimuli

Background: Although interest in crosslinguistic differences in sentence processing has increased in recent years (e.g., Coupé et al., 2019; Jakobi et al., 2025; Siegelman et al., 2022), studies that compare sentence comprehension across languages using matched translation-equivalent stimuli remain relatively rare. In a recent investigation, Chromý et al. (2023) uncovered notable differences in agreement attraction effects between Czech and English. Moreover, a closer analysis of these data revealed a further interesting pattern: native English speakers processed individual words in their language significantly faster than Czech speakers did in theirs. The present study was designed to explore this phenomenon in greater depth.

Method: We created 70 pairs of translation-equivalent sentences (Table 1), matched in length in words between Czech and English. These two languages differ widely in their grammatical characteristics. In contrast to English, Czech has a relatively free word order, it has complex morphology with many declension and conjugation classes, case, number, and gender marking. Generally, Czech is grammatically more complex and informationally denser than English (Sadeniemi et al., 2008).

Table 1: Item example.

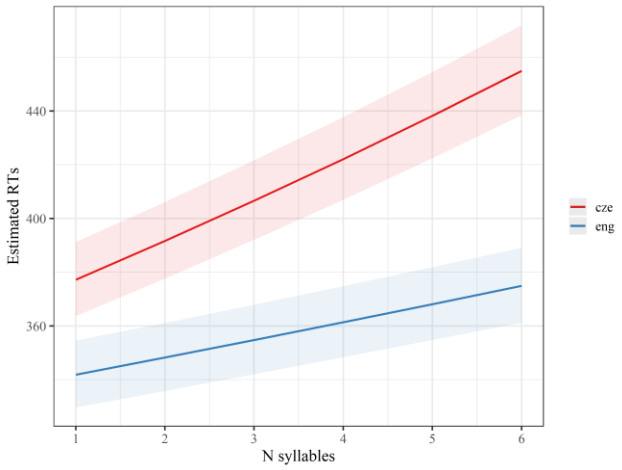
Moje	teta	obvykle	pije	Bordeaux	a	další	francouzská	vína.
My	aunt	usually	drinks	Bordeaux	and	other	French	wines.

The sentences were read by L1 speakers of Czech (N=176) and English (N=176), along with 48 fillers, in a self-paced reading task using a moving-window presentation. After each sentence, participants answered a yes-no comprehension question. Reaction times (RTs) for individual words served as the dependent variable, while the independent variables were language, and word length (in syllables) in Czech.

Results: A linear mixed-effects model with language and word length as fixed effects, and participant and item as random effects, yielded significant main effects of language ($\beta = -0.119$, $SE = 0.026$, $t = -4.598$, $p < .001$) and length ($\beta = 0.039$, $SE = 0.001$, $t = 50.186$, $p < .001$) as well as their interaction ($\beta = -0.02$, $SE = 0.001$, $t = -16.118$, $p < .001$); see Figure 1 for plotted model estimates. In other words, (i) English words were processed faster than Czech words by corresponding L1 speakers, (ii) longer words took more time to process, and (iii) the effect of word length was more pronounced for Czech.

Discussion: The results show robust differences in RTs between English and Czech. We argue that these differences are driven by the greater morphological complexity (i.e. efficiency) of Czech as compared to English. This serves to increase the amount of information encoded per word in the former as compared to the latter (Sadeniemi et al., 2008), leading Czech speakers to expect that the average information processing cost associated with words will be higher than is expected by English speakers, with these different priors being reflected in reading times even when both groups are processing messages that have been encoded in identical numbers of words.

Figure 1: Predicted RTs from the linear mixed-effects model targeting the effects of language and word length.



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